

ALEXANDROS KARAVITIS

+44 7442 717485 Alexis.Karavitis.978@cranfield.ac.uk
Cranfield, UK

PERSONAL STATEMENT

A Mechanical Engineer with a strong passion for motorsport and research driven innovation. Experienced in simulation led design and data driven analysis for composite and additive manufactured structures, 3D CAD and product development. Combines first principles problem solving with practical validation and design for manufacture expertise. Competed in Formula Student as a team leader and engineer before progressing to roles as organiser, judge and scrutineer for Greece's first international competition, demonstrating leadership, precision and resilience under pressure. Skilled in ANSYS, SolidWorks, CATIA and MATLAB, bringing analytical thinking and creativity to high performance engineering environments.

KEY ACHIEVEMENTS

- Led the development of the University of West Attica's first Formula Student vehicle, establishing the team's technical structure, engineering procedures, and project management framework from the ground up. Expanded the team from 6 to over 40 members and delivered the university's debut car to successfully complete the Endurance event at Formula Student East, one of Europe's most demanding competitions.
- Developed a KPI framework using lap time simulation, sensitivity analysis, and DoE to quantify real performance impact of design choices in early concept phases as well as in the R&D processes of a Formula Student racecar.
- Technical Coordinator for the first Formula Student Greece (Formula IHU), contributing to the creation of the event's technical regulations, scrutineering framework, and team assessment processes. Supported the smooth delivery of Greece's first international Formula Student competition involving 500+ participants.
- Developed and validated a simulation-based workflow combining Finite Element Analysis (FEA) with neural networks to predict the laminate properties of 3D-printed structures. This approach reduced reliance on physical testing by enabling accurate material behavior prediction through an AI-driven optimisation process.

EDUCATION

MSc in Advanced Mechanical Engineering, Cranfield University, Cranfield, UK **October 2025 - September 2026**

- **Modules:** Engineering Stress Analysis: Theory and Simulations, Component Design, Computational Fluid Dynamics for Renewable Energy, Artificial Intelligence for Sustainability, Principles of Engineering, Assessing Risk and Failure, Engineering Design and Project Management, Fluid Mechanics and Loading.
- **Best Group Project Presentation Award (May 2026):** Digital Twin and Demonstration of Autonomous Surface Vessels (ASVs).
- **Thesis:** Design of a Fatigue Testing Machine for Elastomer Life Characterization: Simulink Dynamic Modelling , CAD on CATIA V5 and FEA-Based Component Sizing.

Bachelor of Engineering (BEng) Master, University of West Attica, Athens, Greece **September 2018 - July 2025**

- Bachelor of Engineering (BEng) in Mechanical Engineering (5-year Integrated Master).
- **Modules:** Vehicle Dynamics, Computer Aided Mechanical Design I & II, Internal Combustion Engines, Automatic Control Systems, Numerical Methods, Industrial Automation, Applied Statistics, Advanced Materials Technology.
- Grade: 7.26/10.
- **Thesis:** Identification Method for Constitutive Material Parameters in Additive Manufactured Structures using Artificial Intelligence.
- Created an LSTM model in MATLAB for material behaviour prediction in Fused Deposition Modelling (FDM). Implemented Genetic Algorithms (GA) to optimise Finite Element Analysis (FEA) simulations and created an integrated AI-optimisation workflow combining FEA, machine learning, and material science.
- Conducted analysis on Fused Deposition Modelling (FDM) structures deploying classical laminate theory, treating the printed layers as orthotropic plies to model anisotropic mechanical behaviour.

- Finally, applied Finite Element Analysis (FEA) incorporating laminate theory principles to predict stress-strain responses and optimise structural performance. Developed a KPI framework using lap time simulation, vehicle dynamics, and DoE to quantify real performance impact of design choices in early concept phases.

CAREER HISTORY

University of West Attica | Additive Manufacturing Research, Athens, Greece, March 2022 - September 2023 Undergraduate Research

University of West Attica is a public university in Athens focused on applied engineering and technology. The Additive Manufacturing Research Lab specialises in metal and polymer 3D printing.

- Conducted research on Directed Energy Deposition (DED) techniques for metal additive manufacturing in order to associate printing parameters with physical properties.
- Employed Design of Experiment (DOE) methodologies to optimise process parameters and investigated material science properties and manufacturing efficiency improvements.

EXPERIENCE

Formula IHU - Formula Student Greek Competition, Serres, Greece, Technical September 2024 - Present Coordinator

The first ever formula student competition in Greece.

- Organised the first international Formula Student competition in Greece as part of the event team, leading technical, engineering, and scientific departments to help deliver the event with 500+ participants.
- Managed technical regulations, compliance oversight, innovation evaluation, and team assessments and ensured adherence to high engineering standards and formula student regulations.

Formula Student, University of West Attica, Team Manager & Lead Engineer, March 2020 - August 2023 Poseidon Racing Team

A multidisciplinary student engineering team that designs and manufactures Formula Student race cars to compete in international competitions across Europe. The team integrates design, simulation, manufacturing, and testing, while fostering teamwork.

- Managed a 43-member engineering team, overseeing vehicle design and R&D.
- Developed aerodynamic components and structural designs, optimising vehicle efficiency via CFD simulations (Ansys Fluent, ANSA, Meta) and FEA (Ansys Workbench).
- Led manufacturing process planning, including CNC machining, 3D printing, laser cutting, and hydro cutting.
- Created cost analysis, manufacturing documentation, and Bills of Materials (BOMs) of over 200 parts & assemblies.

SKILLS, INTERESTS & EXTRACURRICULAR ACTIVITIES

- **Languages:** Greek (Native), English (Fluent).
- **Technical Skills:** **3D CAD & Mechanical Design** (SolidWorks, CATIA, Autodesk Inventor, AutoCAD) **Finite Element Analysis (FEA)** (Ansys Workbench, Abaqus, SolidWorks), **Computational Fluid Dynamics (CFD)** (Ansys Fluent, ANSA, META), **Programming & AI Applications** (MATLAB, Simulink, C++, Python (Machine Learning), **AI for Engineering** (Neural networks, AI-driven analysis, Claude, Custom Claude Skills, AI-based material modelling), **Project & Cost Management** (Microsoft Project, BOM creation, Cost analysis), **Manufacturing & Prototyping** (CNC Machining, 3D Printing, Laser Cutting, DFM).
- **Certifications:** ANSA/META for Laminated Composites Seminars, BETA CAE Systems (Nov 2021), Introduction to CFD Pre & Post Processing, BETA CAE Systems (Nov 2021), Introduction to Pre & Post Processing with ANSA/META, BETA CAE Systems (Nov 2021), Multi-body Dynamics Analysis with ANSA Kinetics Tool, BETA CAE Systems (Nov 2021).
- **Volunteering Experience:** **Volunteer Firefighter & Rescue Force Member.**
- **Leisure & Interests:** 3D Printing & Prototyping (since 2016, personal printers), Formula 1, Embedded Systems & Automation (Arduino-based IoT, CNC Router development), Philosophy & Literature (Socratic, Hegelian, Aristotelian), Science Fiction (Isaac Asimov, Star Trek), Online Chess (casual & rapid formats), Film Photography (35mm analogue film experimentation).